

**Probability
Fall 2026**

**Homework 2
Due September 14, 2026**

If you use Mathematica you must hand in any Mathematica Notebook that you use to complete the assignment. But I do not generally want electronic copies; just paper please.

1. Let X , Y and Z be random variables and let c be a number. Prove that

$$\text{Var}(cX) = c^2 \text{Var}(X),$$

$$\text{Cov}(cX, Y) = c \text{Cov}(X, Y)$$

and

$$\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z).$$

2. Let X_1 and X_2 be discrete random variables and suppose that a_1 and a_2 are real numbers. Using the joint probability mass function for X_1 and X_2 , show that

$$E[a_1X_1 + a_2X_2] = a_1E[X_1] + a_2E[X_2].$$

And work the following problems from the text, Edition 11:

9, 12, 22, 37, 61 76, 86, along with *16 and *49.

The starred problems are in the back of the book and you **should not** hand these in.

If you have a copy of Edition 10, please let me know so that we can verify that you are working the right problems.

Here is **86** which is not in Edition 10.

86. from Ed. 11 Each new book donated to a library must be processed. Suppose that the time it takes a librarian to process a book has mean 10 minutes and standard deviation 3 minutes. If a librarian has 40 books that must be processed one at a time,

- a. approximate the probability that it will take more than 420 minutes to process all these books.
- b. approximate the probability that at least 25 books will be processed in the first 240 minutes.